

Naomi Rehman

Santa Barbara, CA | <https://naomirehman1008.github.io/> | nrehman@ucsb.edu

EDUCATION

University of California, Santa Barbara <i>PhD in Computer Science — ArchLab, co-advised by Prof. Jonathan Balkind and Prof. Timothy Sherwood</i>	<i>Expected 2030</i>
University of California, Santa Cruz <i>B.S. Computer Engineering, Highest Honors GPA 3.7</i> <i>B.A. Computer Science GPA: 3.8</i>	<i>2020 – 2025</i>

PUBLICATIONS AND PRESENTATIONS

- “SIMT-Step Execution: A Flexible Operational Semantics for GPU Subgroup Behavior”, Zheyuan Chen, **Naomi Rehman**, Guido Martinez, Tyler Sorensen, PLDI 2026
- “I2P: Incorrect Path Prediction for Power Throttling Speculative Execution”, **Naomi Rehman**, B.S. Senior Thesis, UC Santa Cruz, 2025 (Steck Family Award for Best Senior Thesis at UCSC)
- “I2P: Incorrect Path Prediction”, **Naomi Rehman**, YArch Workshop, ASPLOS 2025 (lightning talk and poster)
- “Confidence: a Light-Weight Instruction Prefetcher”, **Naomi Rehman**, UC Santa Cruz ACM Conf 2024 (presentation and poster)

AWARDS AND FELLOWSHIPS

• NSF Graduate Research Fellowship	<i>2026-2030</i>
• UCSB CS Departmental Fellowship	<i>2025-2027</i>
• CRA Outstanding Undergraduate Researcher Award, Honorable Mention	<i>2025</i>
• Steck Family Award for Best Senior Thesis at UCSC	<i>2025</i>
• Chancellor's Award for Senior Thesis	<i>2025</i>
• Dean's Award for Senior Thesis	<i>2025</i>

SKILLS

- **AI:** PyTorch, CUDA, vLLM, Timeloop, Triton
- **Software:** C/C++, Python, Pandas
- **Architecture:** CPU & GPU Architecture, Cycle-Accurate Simulation (Scarab), GPU Assembly, Verilog/SystemVerilog

WORK EXPERIENCE

PhD Researcher (Architecture) UC Santa Barbara, Santa Barbara, CA	<i>2025 – Present</i>
• Researching a hardware-software codesigned system for energy efficient Zeroth-Order training, enabling fine-tuning of AI models on edge devices. ◦ Implemented several variants of 2D systolic arrays and evaluated their power consumption and area, leveraging SystemVerilog, Yosys, and OpenROAD for design, synthesis, and layout. ◦ Extended a Timeloop-based evaluation flow by writing new config files and adding post-processing to model new architecture/algorithm configurations for LLM training on a 2D systolic-array-based accelerator. ◦ Leading a team of three undergraduate students. ◦ <i>Paper “eZO” in preparation for submission.</i> • Built a fast evaluation framework for a near-data processing architecture, combining analytical models and numeric optimizers to rapidly estimate power, performance, and area (PPA).	
GPU Architecture Intern NVIDIA, Santa Clara, CA	<i>Summers 2024, 2025, 2026</i>
• 2026: Designed, prototyped, and evaluated a potential new architecture feature in a cycle-approximate simulator (C++). Built CUDA software leveraging a separate new feature to improve perf/watt, achieving 1.9% improvement. • 2025: Developed functional tests for a new GPU architectural feature using C++ and GPU assembly. Restructured existing functional tests to reduce redundancy, yielding up to 67x test speedup (3x on average). • 2024: Developed new testing infrastructure for GPU architecture using Python, C++, and GPU assembly, identifying bugs overlooked by existing systems.	

Advised by Prof. Heiner Litz

- Developed a hardware prediction mechanism for detecting off-path cycles (cycles following mispredictions) in high-performance CPUs, implemented in the Scarab cycle-accurate CPU simulator.
- Achieved 86.8% average accuracy predicting whether cycles were on or off-path for SPEC2017 workloads.
- *Undergraduate thesis awarded Steck Family Award, the highest undergraduate honor at UCSC.*

Advised by Prof. Tyler Sorensen

- Built a cross-platform GPU testing tool to expose workgroup-level forward-progress properties, including a transpiler to convert concurrent pseudo-code to WGSL and WebGPU for running conformance tests.
- Tested the subgroup semantics of several GPU platforms with the Amber testing framework.
- Identified and reported a WebGPU-triggered DoS vulnerability causing system crashes on Apple Silicon and Samsung systems.
- *Paper “SIMT-Step” published in PLDI 2026.*

- Engineered a cost-effective automated mechatronic system to precisely and consistently deliver machining fluid during the manufacturing of x-ray telescope mirrors, reducing active labor per mirror by 20%.

PROJECTS

- Realized theoretical performance gains of FlexiDepth (Luo et al.) by adding an early-exit decode path based on router decisions to the vLLM Triton attention kernel, achieving ~19% decode throughput improvement over Luo et al.'s implementation and ~15% over vanilla Llama-3 8B at 12,000-token context, batch size 16 on an A6000 GPU.

- Modified the vLLM block tables and management to support a memory-efficient transformer variant, achieving up to ~14% token throughput improvement by reducing KV cache footprint by 25%.

- Implemented the Best-Offset Data Prefetcher (Michaud, 2016) in the Scarab CPU simulator and augmented the score table with Bloom filters to reduce useless prefetches, improving IPC by ~3% on average over vanilla BOP.

- Implemented a hybrid CNN-GNN surrogate model for fluid flow simulation with PyTorch, achieving an average coefficient of determination (R^2) of 0.94.

- Implemented a FUSE-based file system in C with inodes, an inode table, and multi-level indirect block addressing, capable of cloning and building SQLite on Linux.

RELEVANT COURSEWORK

Computer Architecture (RISC-V, C++) · Digital Design (Verilog) · ASIC Design (SystemVerilog) · High Speed IC Design · Parallel Programming (C++, Python) · Operating Systems (C) · Distributed Systems · Foundation Models (PyTorch) · AI for Science (PyTorch) · Data Structures and Algorithms (C, C++)